

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

x	y
1	5
2	7
3	9
4	11

The table above shows some pairs of x values and y values. Which of the following equations could represent the relationship between x and y ?

Answer

- A. $y = 2x + 3$
- B. $y = 3x - 2$
- C. $y = 4x - 1$
- D. $y = 5x$

Correct Answer: A

Rationale

Choice A is correct. Each of the choices is a linear equation in the form $y = mx + b$, where m and b are constants. In this equation, m represents the change in y for each increase in x by 1. From the table, it can be determined that the value of y increases by 2 for each increase in x by 1. In other words, for the pairs of x and y in the given table, $m = 2$. The value of b can be found by substituting the values of x and y from any row of the table and substituting the value of m into the equation $y = mx + b$ and then solving for b. For example, using $x = 1$, $y = 5$, and $m = 2$ yields $5 = 2(1) + b$. Solving for b yields $b = 3$. Therefore, the equation $y = 2x + 3$ could represent the relationship between x and y in the given table.

Alternatively, if an equation represents the relationship between x and y, then when each pair of x and y from the table is substituted into the equation, the result will be a true statement. Of the equations given, the equation $y = 2x + 3$ in choice A is the only equation that results in a true statement when each of the pairs of x and y are substituted into the equation.

Choices B, C, and D are incorrect because when at least one pair of x and y from the table is substituted into the equations given in these choices, the result is a false statement. For example, when the pair $x = 4$ and $y = 11$ is substituted into the equation in choice B, the result is $11 = 3(4) - 2$, or $11 = 10$, which is false.